Introduction

A Synthetic Study of Checklist-Assisted Review

Reviewers inspect short reports to identify reporting omissions.

This demonstration examines whether a checklist improves omission detection.

RQ1: Does checklist access increase the number of omissions detected?

The aim was to compare checklist-assisted review with unaided review.

Study Design

Twelve volunteer graduate students were recruited through a campus email.

Participants were randomly assigned to two groups of six in a between-subjects design.

Both groups reviewed the same two-page report for 15 minutes and listed omissions.

The checklist group received a printed checklist; the control group reviewed without it.

Methods

Participants gave consent, received a two-minute tutorial, completed the task, then rated confidence.

This synthetic demonstration involved no real human participants.

The interface recorded each submitted omission and its timestamp during the task.

After the task, participants rated confidence on a scale from 1 to 5.

For each participant, duplicate omission labels were removed and remaining labels were counted.

We compared omission counts between groups using a two-sided Mann-Whitney U test.

Table 1. Data sources and collection times in the synthetic study.

Data	Collected when	Unit
Omission log	During task	Submission
Confidence rating	After task	Participant

Results

Median omission counts were 8 for checklist and 5 for control; U = 5, p = 0.041.

The checklist group detected more omissions than the control group in this task.

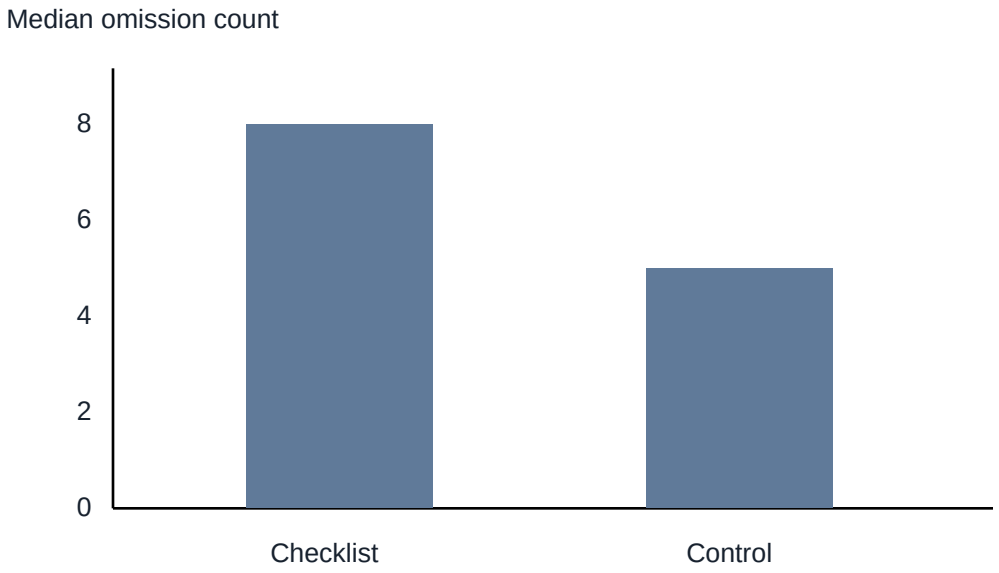

Figure 1. Median omission counts by condition. Bars encode medians; no error bars are shown.

Discussion

The authors interpret this pattern as possible support for structured review aids.

This demonstration illustrates a traceable workflow for evaluating a review checklist.

The small sample and single report limit generalization.